

22 . CANDIDATE-ELIMINATION ALGORITHM

Aim

To implement the Candidate-Elimination algorithm and determine the specific boundary (S) and general boundary (G) of the version space for a given training dataset.

Algorithm + Procedure

1. Read the training examples from the CSV file.
2. Initialize the specific boundary S with the first positive example.
3. Initialize the general boundary G with the most general hypothesis containing ? in every attribute.
4. For every training example, check whether it is positive or negative.
5. For a positive example, generalize S wherever its attribute values differ from the example.
6. Remove hypotheses from G that are not consistent with the positive example.
7. For a negative example, specialize the hypotheses in G.
8. Generate hypotheses that are consistent with the specific boundary S.
9. Continue until all training examples are processed.
10. Display the final S and G boundaries.

Code

```
import pandas as pd

def more_general(h1, h2):
    return all(x == '?' or x == y for x, y in zip(h1, h2))

def candidate_elimination(X, y):
    n = X.shape[1]
    S = [X[0].copy()]
    G = [['?'] * n]
    for i in range(len(X)):
        x = X[i]
        if y[i].lower() == 'yes':
            for j in range(n):
                if S[0][j] != x[j]:
```

```

        S[0][j] = '?'

    G = [g for g in G if more_general(g, x)]

else:

    new_G = []

    for g in G:

        for j in range(n):

            if g[j] == '?' and S[0][j] != x[j]:

                h = g.copy()

                h[j] = S[0][j]

                new_G.append(h)

    G = new_G

return S, G

data = pd.read_csv('enjoysport.csv')
X = data.iloc[:, :-1].values
y = data.iloc[:, -1].values
S, G = candidate_elimination(X, y)
print("Final S:", S)
print("Final G:", G)

```

Output:

```

Final S: [array(['Sunny', 'Warm', '?', 'Strong', '?', '?'], dtype=object)]
Final G: [['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?']]

```

Result

The Candidate-Elimination algorithm was successfully implemented and the final specific and general boundaries of the version space were obtained.